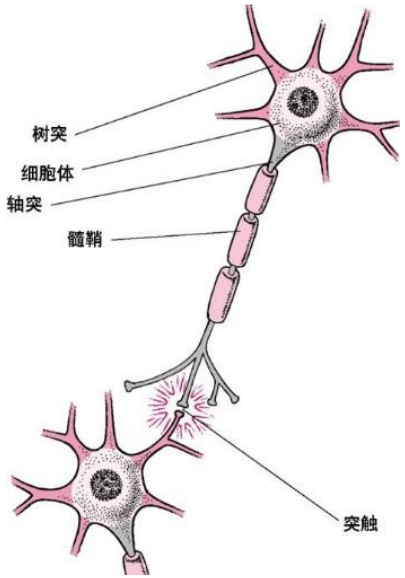
Spiking Neuronal Networks

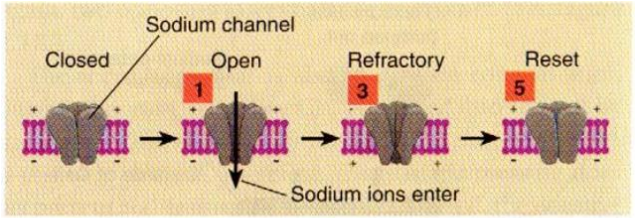
PMI group meeting KWT

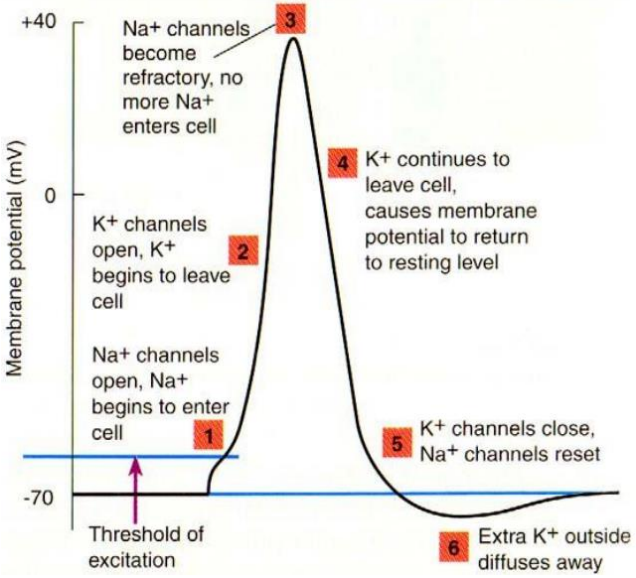
LYH Mar 25, 2023

Neuron Models

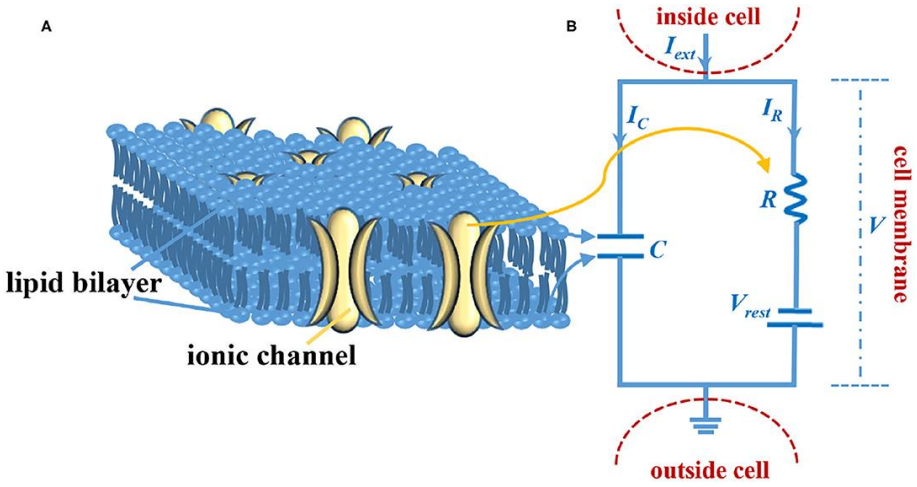
Biological neuron







Leaky Integrate and Fire Model



$$\tau \frac{d}{dt} V(t) = V_{rest} - V(t) + RI(t)$$

$$\tau = RC$$

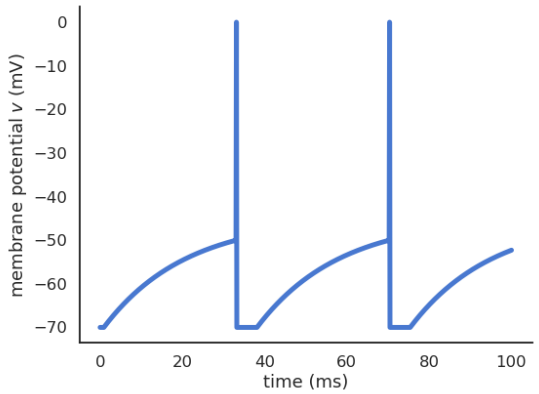
$V(t') = V_\theta \rightarrow$ Fire

V_θ: threshold

$V(t) = V_{reset}$ for $t \in (t', t' + t_r)$

$V_{reset} = V_{rest}$

t_r refractory period

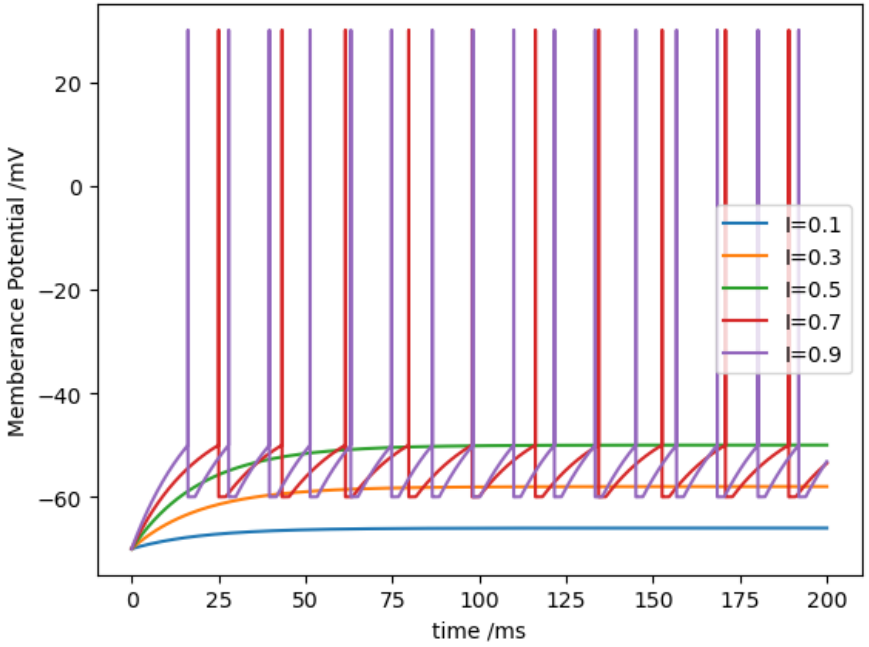


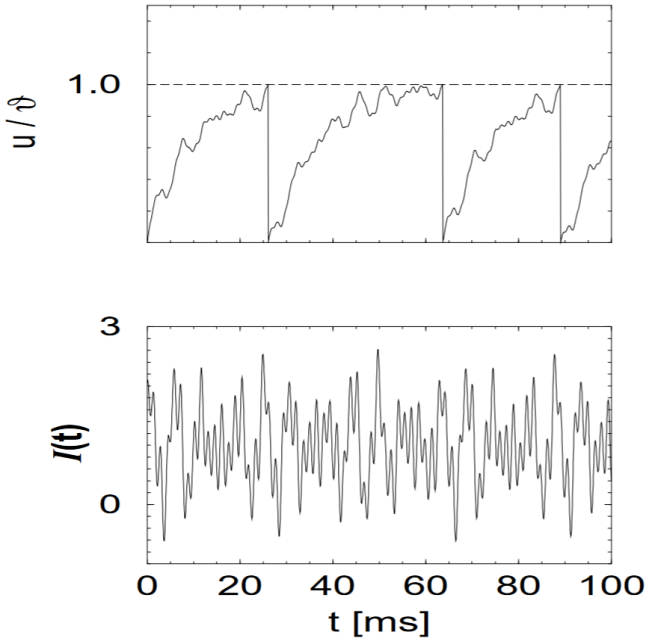
$$\tau \frac{d}{dt} V(t) = V_{rest} - V(t) + RI(t)$$

- **Constant input**

$$V(t) = V_{rest} + RI_0 \left(1 - \exp\left(-\frac{t}{\tau}\right)\right)$$

Firing rate $\quad r = \dfrac{1}{t_r + \tau \ln\left(1 + \dfrac{V_\theta - V_{rest}}{R(I - I_c)}\right)}$





- **Time-dependent input**

$$V(t) = V_{rest} + \frac{R}{\tau} \int_0^\infty \exp\left(-\frac{s}{\tau}\right) I(t - s)\, \mathrm{d}s$$

reset

$$I_r = -q_r \sum \delta(t - t') = -C(V_\theta - V_r) \sum \delta(t - t')$$

$s \equiv t - t'$

$S(t - t') \equiv \sum \delta(t - t')$

$\eta(s) = (V_r - V_\theta) \exp\left(-\frac{s}{\tau}\right) \qquad \kappa(s) = \frac{1}{C} \exp\left(-\frac{s}{\tau}\right)$

$$V(t) = \int_0^\infty \eta(s)\, S(t - s)\, \mathrm{d}s + \int_0^\infty \kappa(s)\, I(t - s)\, \mathrm{d}s$$

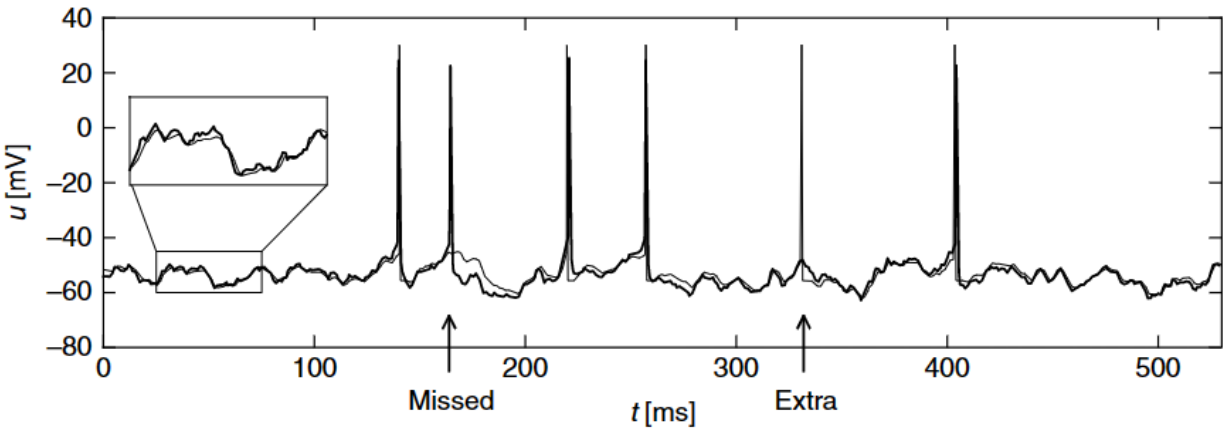
total current $I + I_r$

$$V(t) = \sum (V_r - V_\theta) \exp\left(-\frac{s}{\tau}\right) S(t - s) + \frac{R}{\tau} \int_0^\infty \exp\left(-\frac{s}{\tau}\right) I(t - s) \mathrm{d}s + V_{rest}$$

Threshold Adaptation

$$\tau_0 \frac{d}{dt} V_\theta = \theta - V_\theta + a \sum_k \delta(t - t'^k)$$

θ resting value of V_θ



Synaptic Input

CUBA current based

$$I(t) = \sum_i w_i \sum_{t'} \mathrm{psc}(t - t_i')$$

$\mathrm{psc}(t)$ postsynaptic current

COBA conductance based

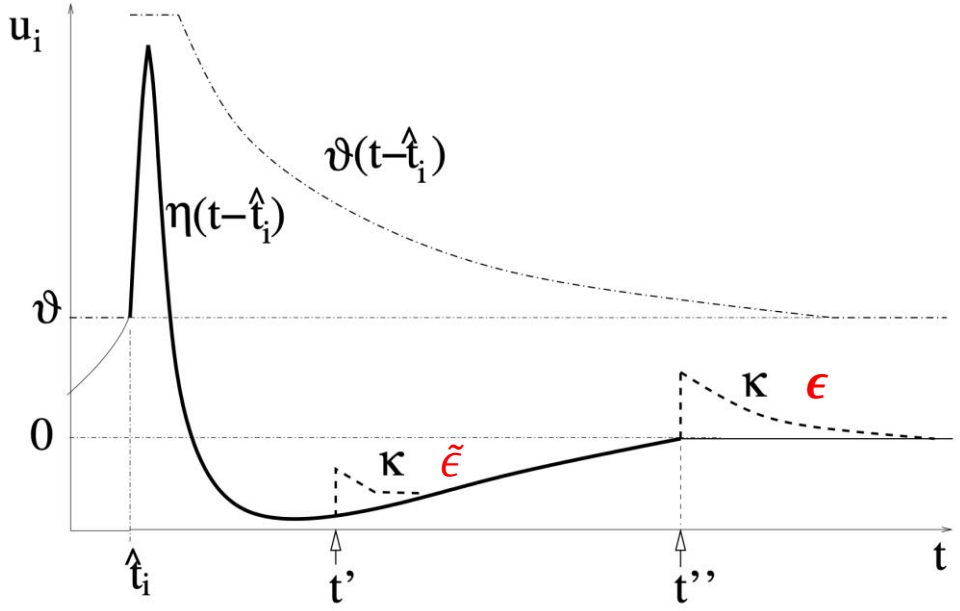
$$I(t) = \sum_i w_i \sum_{t'} g_{syn}(t - t_i')(V - V_{syn})$$

V_{syn} the reversal potential of the synapse

Spike-Response Model

$$V(t) = \sum_i \eta(t-\hat{t}_i) + \sum_j w_{ij} \sum_s \epsilon(t-\hat{t}_i, s) + \int_0^\infty \tilde{\epsilon}\,(t-\hat{t}_i, s)\, I^{ext}(t-s)\, \mathrm{d}s$$

Gerstner, W. (2008). Spike-response model. *Scholarpedia*,*3* (12), 1343.



$$V(t) = \sum_i \eta(t-\hat{t}_i) + \int_0^\infty \kappa(t-\hat{t}, s)\, I(t-s)\, \mathrm{d}s$$

$$\kappa(t-\hat{t}, s) = a_0 \exp\left(-\int_0^s \frac{\mathrm{d}t'}{\tau(t-\hat{t})}\right)$$

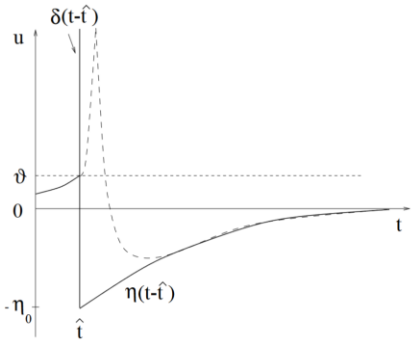
Gerstner, W. et al. (2014). Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition

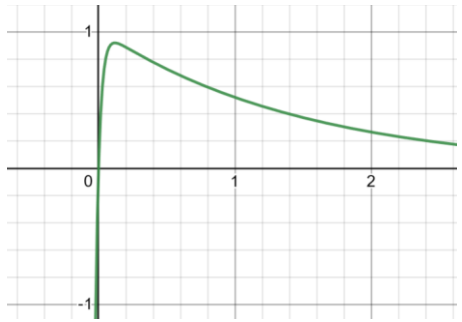
SRM0

$$V(t) = \eta_i(t-\hat{t}) + \sum_j w_{ij} \sum_s \epsilon_0(s)$$

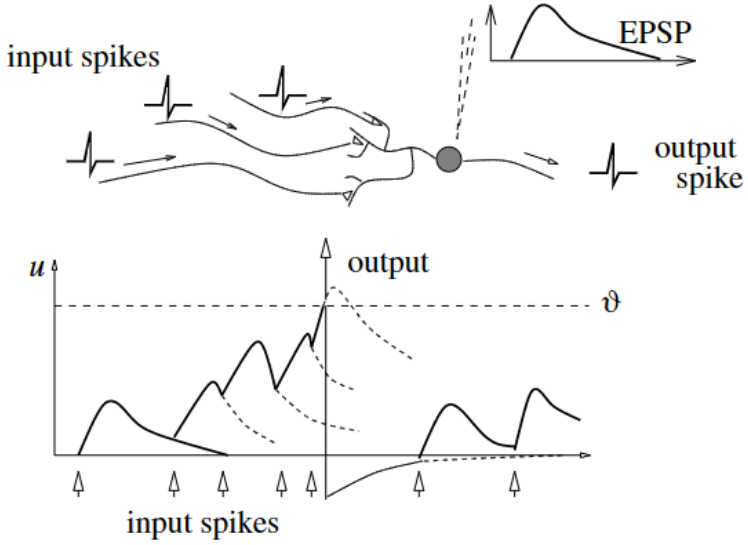
$$\eta(t-\hat{t}) = \delta(t-\hat{t}) - \eta_0 \exp\left(-\frac{t-\hat{t}}{\tau}\right)$$

$$\epsilon_0(s) = \frac{1}{1-\frac{\tau_c}{\tau_m}}\left(\exp\left(-\frac{s}{\tau_m}\right) - \exp\left(-\frac{s}{\tau_c}\right)\right)$$

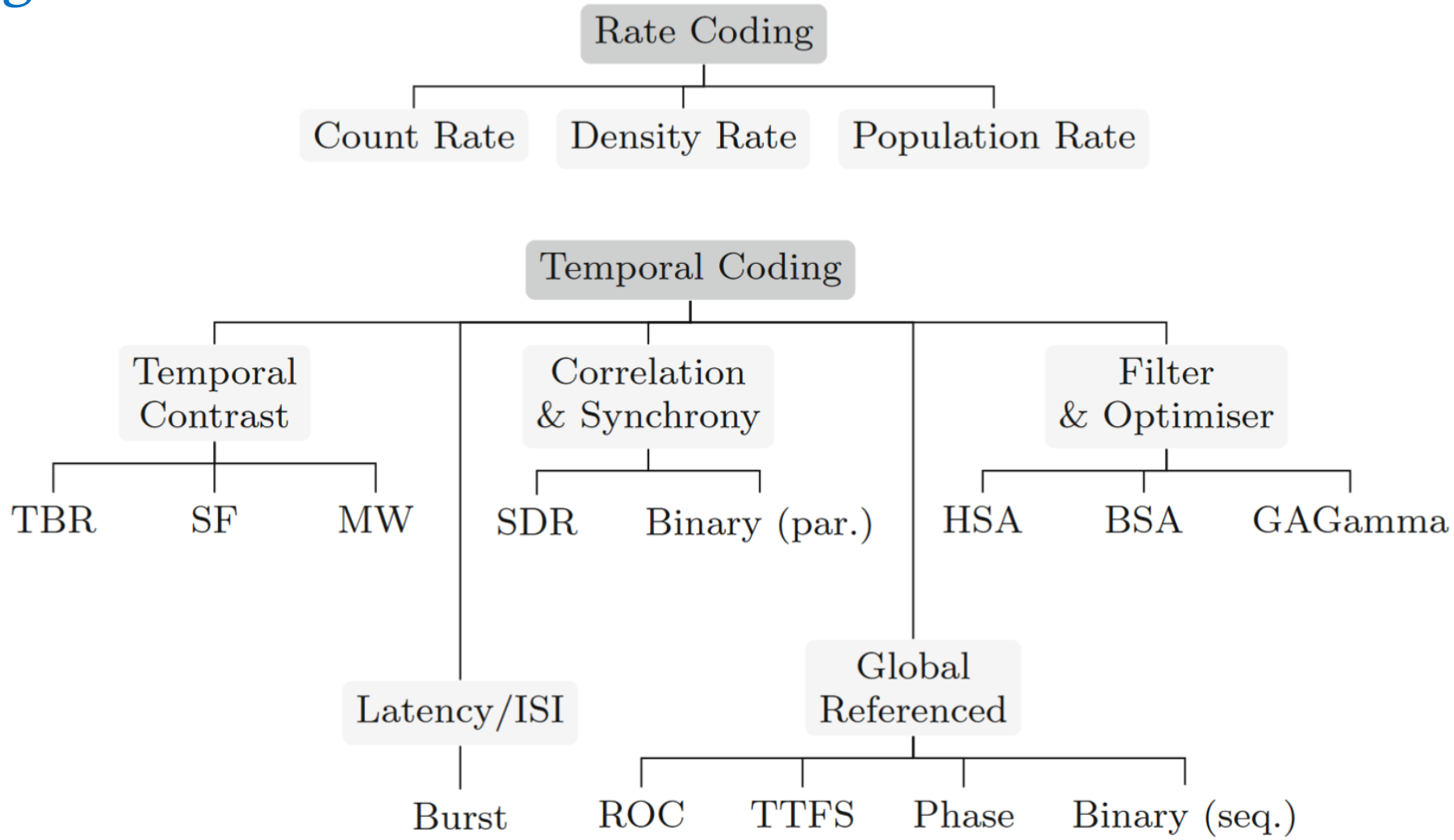




$\tau_m = 1.5, \qquad \tau_c = 0.03$

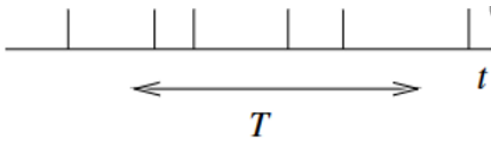


Nerual coding



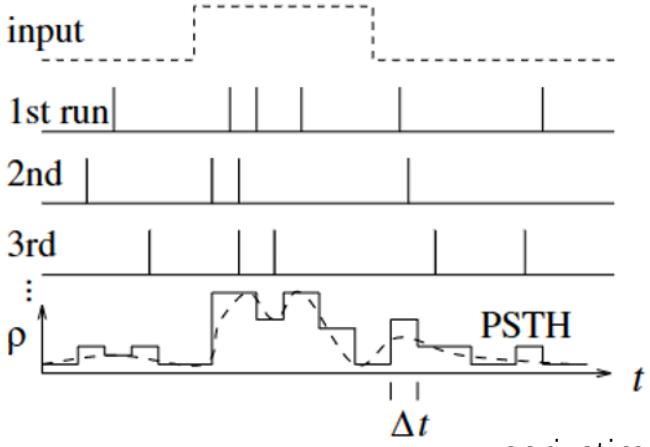
Frequency Coding

- average over time



$$\nu = \frac{n_{sp}}{T}$$

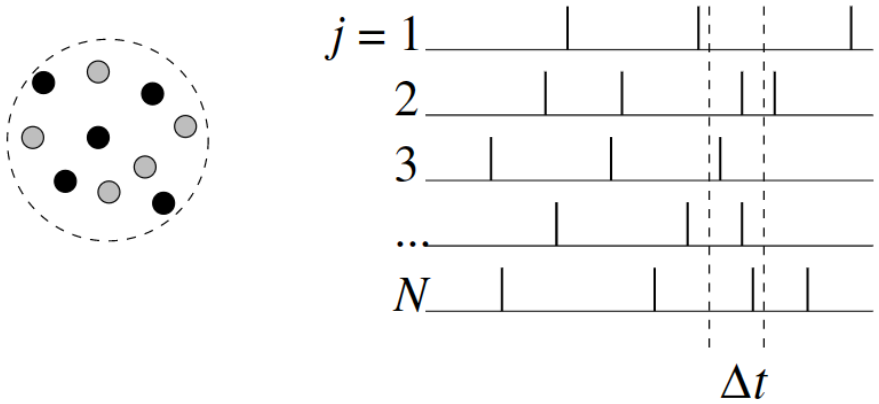
- average over several runs



peri-stimulus-time histogram
周围刺激时间直方图

$$\rho(t) = \frac{1}{\Delta t}\frac{n_K(t; t+\Delta t)}{K}$$

- average over several neurons

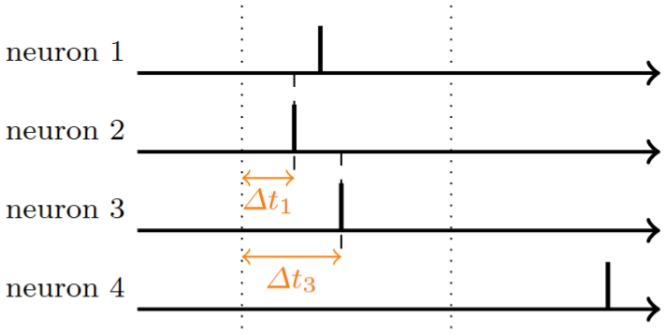


$$A(t) = \frac{1}{\Delta t}\frac{n_{act}(t; t+\Delta t)}{N}$$

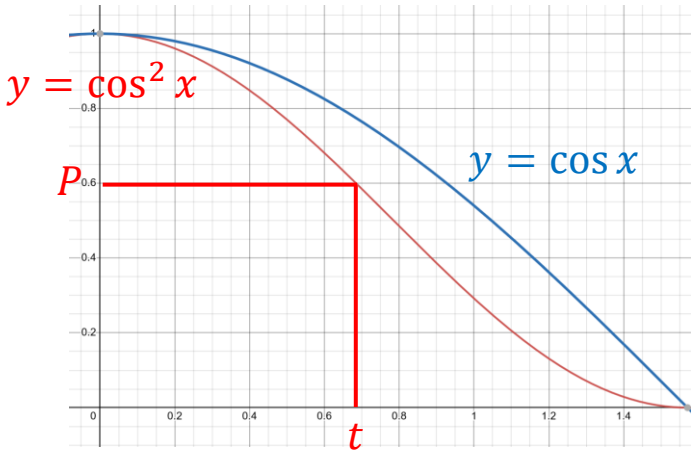
- reflect changes in the stimulus conditions rapidly
- real populations always have a certain degree of heterogeneity

Tempotral Coding

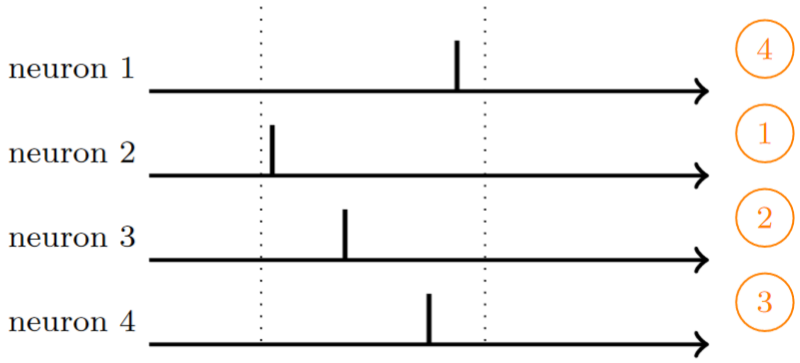
- Time-to-first-spike



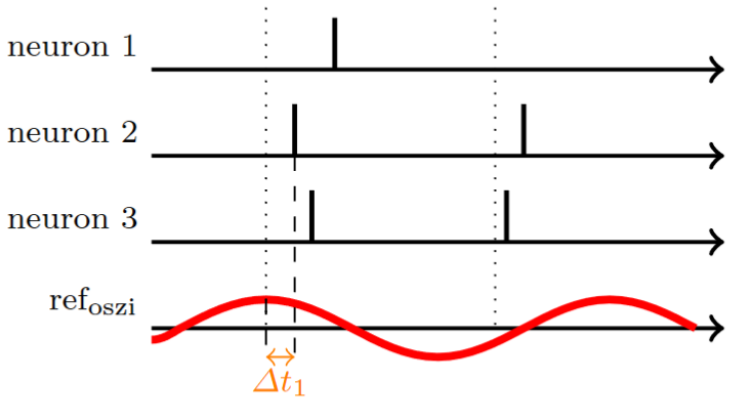
(b) TTFS coding



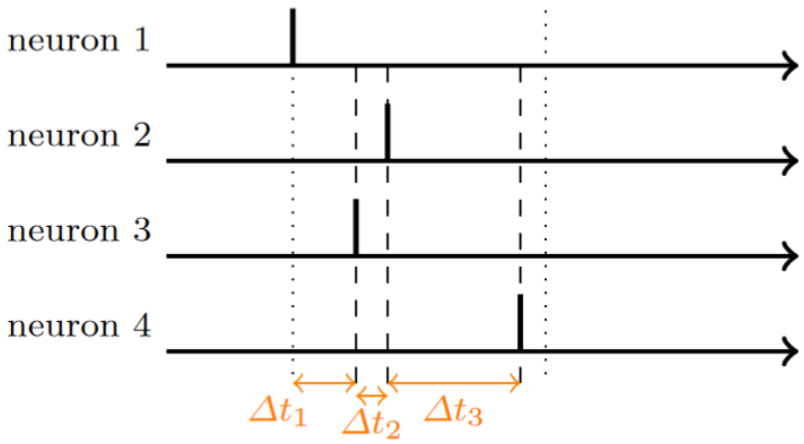
- Rank order Coding



- Phase Coding



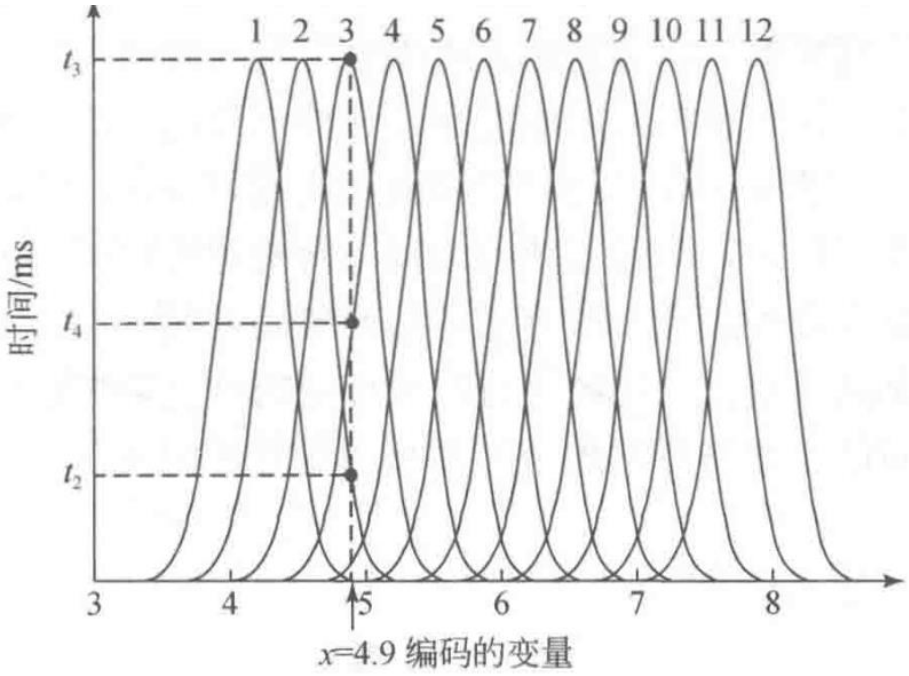
- Latency Coding



Population Coding

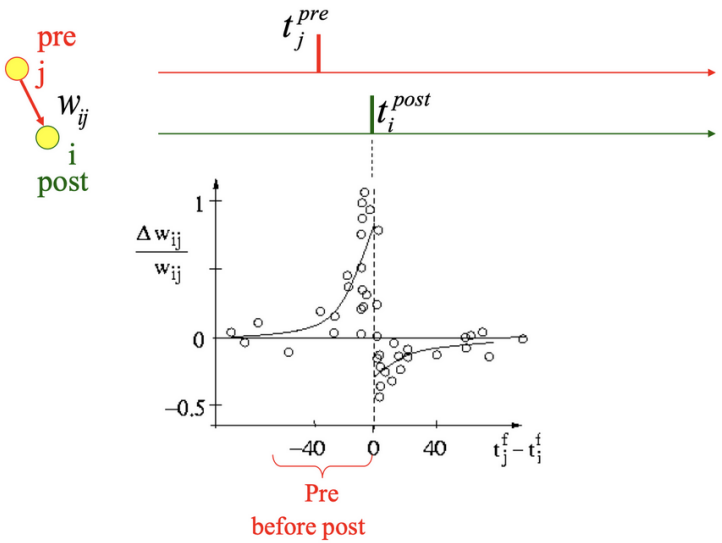
$$\mu = I_{\min} + \frac{2i-3}{2}\frac{I_{\max} - I_{\min}}{m-2}$$

$$\sigma = \frac{1}{\beta}\frac{I_{\max} - I_{\min}}{m-2}$$

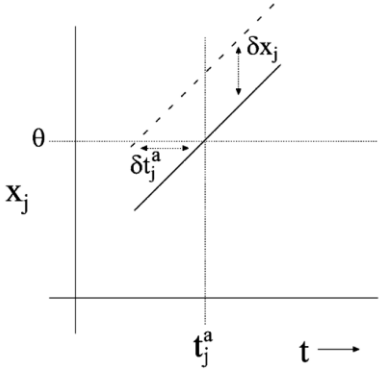


Learning algorithm

- STDP (Spike Timing Dependent Plasticity)



$$\Delta w_{ij} = \begin{cases} a^+ \exp\left(\frac{t_j - t_i}{\tau^+}\right), & t_j - t_i > 0; \\ a^- \exp\left(\frac{t_i - t_j}{\tau^-}\right), & t_j - t_i \leqslant 0. \end{cases}$$



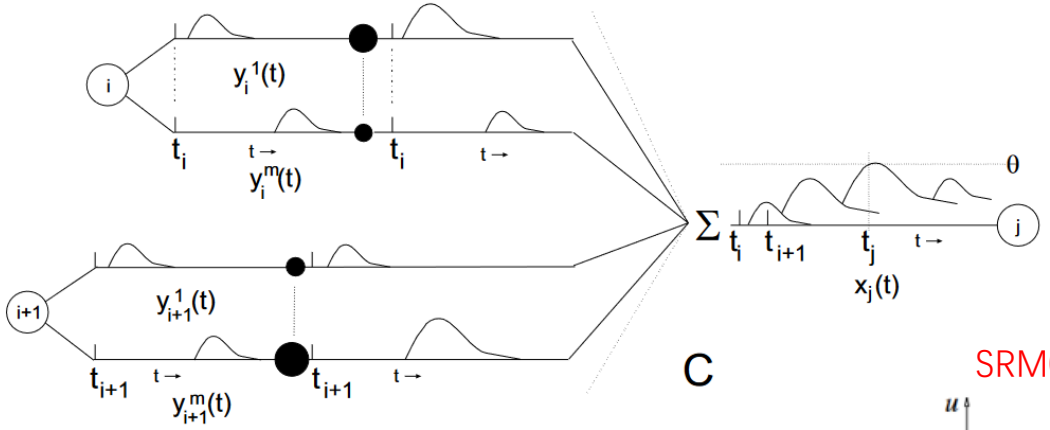
- SpikeProp

$$\varepsilon(t) = \frac{t}{\tau} e^{1-\frac{t}{\tau}}$$

$$y_i^k(t) = \varepsilon(t - t_i - d^k)$$

$$y_i(t) = \Sigma_k^m y_i^k(t)$$

$$x_j(t) = \Sigma_i \Sigma_k w_{ij}^k y_i^k(t)$$



C

$\{ t_j^d \}$

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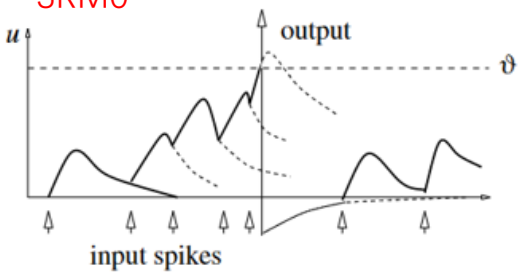


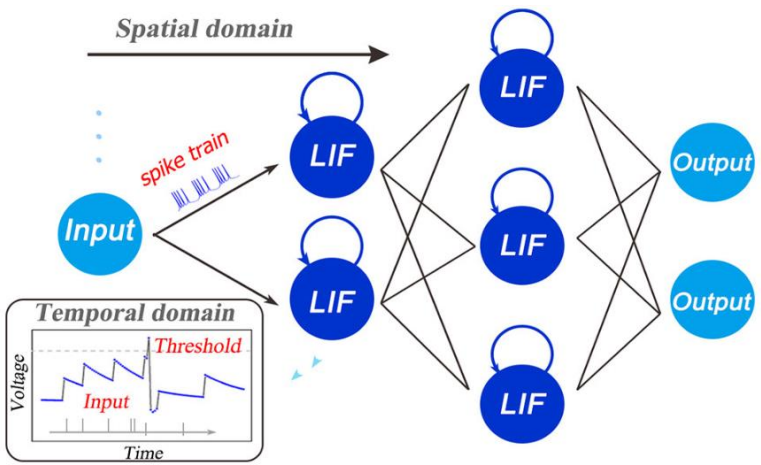
$$E = \frac{1}{2} \Sigma_j (t_j - t_j^d)^2$$

$$\Delta w_{ij}^k = -\eta \frac{\partial E}{\partial w_{ij}^k} = -\eta \frac{\partial E}{\partial t_j} \frac{\partial t_j}{\partial w_{ij}^k} = -\eta (t_j - t_j^d) \frac{\partial t_j}{\partial w_{ij}^k} = -\eta (t_j - t_j^d) \frac{\partial t_j}{\partial x_j(t)}\Big|_{t=t_j} \frac{\partial x_j(t)}{\partial w_{ij}^k}\Big|_{t=t_j}$$

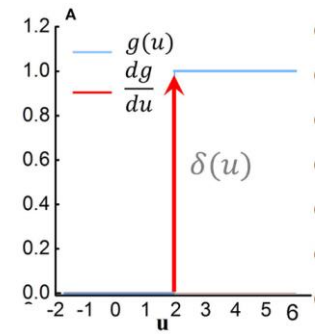
linear hypothesis $\quad \alpha = -\frac{\partial x_j(t)}{\partial t}\Big|_{t=t_j}$

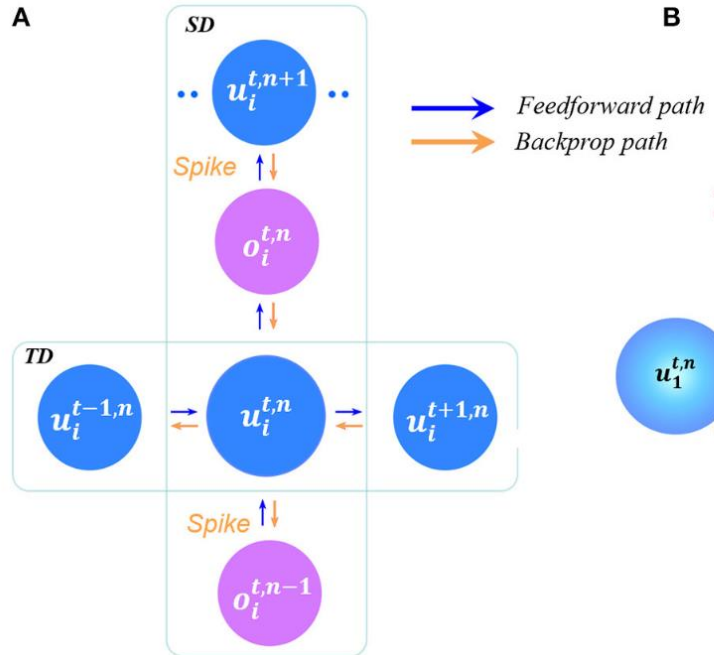
$$\frac{\partial t_j}{\partial x_j(t_j)} = \frac{-1}{\alpha} = \frac{1}{\partial x_j(t_j)/\partial t_j} = \frac{1}{\Sigma_{i,l} w_{ij}^l (\partial y_i^l(t_j)/\partial t_j)}$$

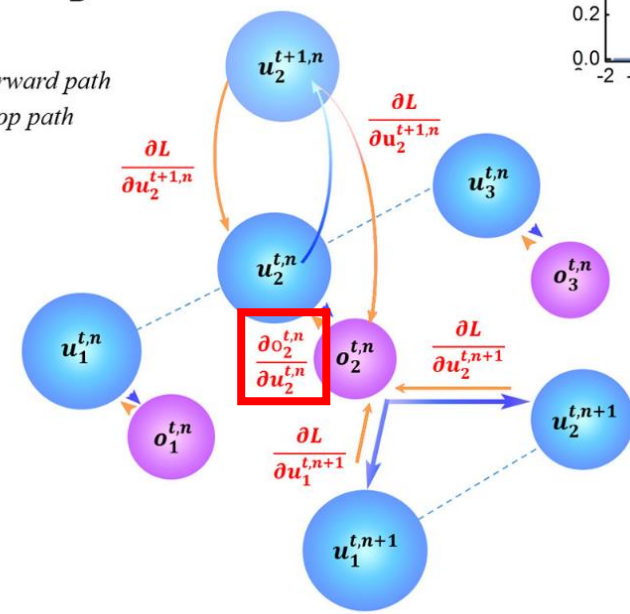
- STBP



Loss $L = \frac{1}{2S}\sum_{s=1}^{S} \| y_s - \frac{1}{T}\sum_{t=1}^{T} o_s^{t,N} \|_2^2$







Simplified LIF model $u(t) = u(t_{i-1})e^{\frac{t_{i-1}-t}{\tau}} + I(t)$

$$x_i^{t+1,n} = \sum_{j=1}^{l(n-1)} w_{ij}^n o_j^{t+1,n-1}$$

$$u_i^{t+1,n} = u_i^{t,n} f(o_i^{t,n}) + x_i^{t+1,n} + b_i^n$$

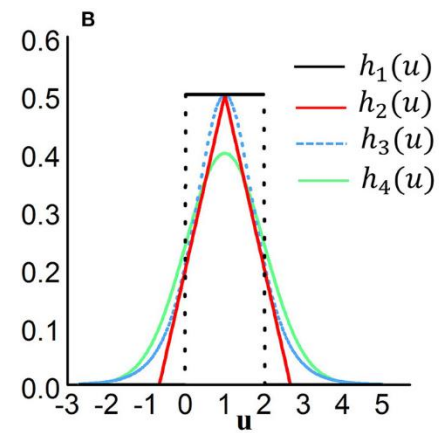
$$O_i^{t+1,n} = g(u_i^{t+1,n})$$

$$f(x) = \tau e^{-\frac{x}{\tau}} \ , \ g(x) = \begin{cases} 1 & x \geq V_{th} \\ 0 & x < V_{th} \end{cases}$$

$$f(o_i^{t,n}) = \begin{cases} \tau & o_i^{t,n} = 0 \\ 0 & o_i^{t,n} = 1 \end{cases}$$

$$h_1(u) = \frac{1}{a_1} sign(|u - V_{th}| < \frac{a_1}{2})$$

$$h_2(u) = (\frac{\sqrt{a_2}}{2} - \frac{a_2}{4}|u - V_{th}|) sign(\frac{2}{\sqrt{a_2}} - |u - V_{th}|)$$

$$h_3(u) = \frac{1}{a_3} \frac{e^{\frac{V_t h - u}{a_3}}}{(1 + e^{\frac{V_t h - u}{a_3}})^2}$$

$$h_4(u) = \frac{1}{\sqrt{2\pi a_4}} e^{\frac{(u - V_t h)^2}{a_4}}$$